
HoloChain-Cert: Holographic Watermarking Meets STARK-Based Provenance for Text

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Abstract

1 We address the open challenge of end-to-end provenance for language-model text
2 that remains reliable on brief quotations, cryptographically auditable, energy-aware,
3 and multilingual. Existing watermarks satisfy at most two of these goals, faltering
4 on < 40 -token snippets, requiring trusted setups, ignoring chain-of-custody, or
5 inflating latency and power draw. HoloChain-Cert unifies six elements: (i) a
6 tri-granular Moiré lattice embedding rateless-coded bits every 2-32 tokens; (ii)
7 a CRS-free STARK circuit that binds each digest to an append-only Poseidon-
8 Merkle ledger; (iii) a verifiable hand-off protocol for multi-agent editing chains;
9 (iv) an NVML-driven policy that disables costly lattice layers under peak load;
10 (v) a 32-language normalising-flow adapter; and (vi) a public opt-out flag. We
11 implement the full stack in CUDA, Cairo, and Rust and benchmark on WaterBench-
12 2.2, ChainBench-1.0, and LowRes-Mix against five competitive baselines. The
13 provenance layer achieves 100 % lineage accuracy with 48 B proofs verified
14 in 1 ms, but short-excerpt detection reaches only 4.2 % TPR at 10^{-6} FPR, and
15 adaptive energy savings average 17 % with a 6 pp low-resource gap. We analyse
16 failure modes, release reproducibility artefacts, and outline concrete fixes toward
17 dependable, sustainable watermarking.

18 1 Introduction

19 Large language models (LLMs) now generate prose that is reposted, remixed, and paraphrased at Inter-
20 net speed, raising urgent questions of provenance, authorship, and environmental cost. Watermarking-
21 the embedding of imperceptible statistical signals-has become the leading technical mitigation,
22 enabling detectors to flag machine-generated text without harming fluency. Nevertheless, three
23 stubborn gaps remain. First, today’s lexical or semantic watermarks lose statistical power on excerpts
24 shorter than roughly forty tokens or after structurally disruptive paraphrases such as sentence fusion
25 and faceted summarisation. Second, legal or journalistic workflows demand more than a binary
26 detector: they require cryptographically auditable records linking each edit to a responsible agent.
27 Existing cryptographic overlays rely on Groth16 zk-SNARKs and a trusted common-reference string
28 (CRS), complicating key rotation and undermining public verifiability. Third, always-on watermark
29 encoders add latency and GPU power draw, yet the environmental cost of such overhead has scarcely
30 been quantified, let alone mitigated.

31 We introduce HoloChain-Cert, a system that tackles all three gaps by weaving together ideas from
32 distribution-preserving watermarking, token-specific logit shaping, publicly verifiable schemes, and
33 the watermark-throughput trade-off theorem. Concretely, HoloChain-Cert couples a holographic
34 Moiré lattice with a CRS-free STARK-backed ledger, adds an NVML-based energy-adaptive switch,
35 and deploys a language-universal normalising-flow mesh so that one key serves thirty-two languages
36 and code-switch sentences.

1.1 Key contributions of HoloChain-Cert

- **Tri-granular Moiré lattice:** Under ideal conditions, reconstructs payload bits with $\leq 0.3\%$ BER on windows as short as twenty tokens.
- **CRS-free STARK provenance:** First CRS-free STARK circuit for watermark provenance; 48 B proofs verify in ≈ 1 ms in commodity browsers.
- **Append-only ledger:** Poseidon-Merkle ledger that records every hand-off and survives ISA-MAP key rotation.
- **Energy-adaptive policy:** NVML-based control that disables macro-grids when utilisation exceeds 80 %, targeting 45 % energy savings.
- **Multilingual flow adapter:** 128-dimensional normalising-flow adapter trained on thirty-two languages plus code-switching, aiming for ≤ 4 pp Δ TPR relative to English.
- **Reproducible implementation:** Open-source CUDA, Cairo, and Rust stack with YAML-controlled hyper-parameters, golden-vector tests, and hard assertions that abort runs when success criteria are unmet.

By integrating holographic embedding, zero-knowledge proofs, and energy awareness, HoloChain-Cert aspires to shift watermarking from single-shot detection toward court-ready provenance. Our first evaluation cycle validates the cryptographic layer but exposes weaknesses in short-excerpt robustness and multilingual energy savings, guiding the next research iteration.

2 Related Work

Vocabulary-partition watermarks bias token sampling using secret green lists. They achieve high detectability on long passages but falter on excerpts under fifty tokens and under paraphrase stress. Token-specific logit shaping strengthens per-token signal at the cost of greater sampling bias, while semantic-invariant schemes derive logits from contextual meaning to boost synonym resilience, yet still assume at least sixty-four observed tokens. Retrieval-based defences store every generated output and flag near duplicates, which scales poorly and offers no cryptographic audit trail.

Publicly verifiable schemes decouple generation and detection networks, allowing anyone to audit outputs without a secret key; however, they rely on Groth16, whose trusted setup complicates key revocation. Distribution-preserving watermarks minimise perplexity shifts and avoid green-list artefacts, but neither attribute edits nor record lineage. Energy aspects have received minimal attention; a recent no-go theorem formalises a trade-off between watermark strength and speculative sampling speed without proposing adaptive mechanisms.

Beyond text, ProMark embeds causal watermarks into diffusion-model training data to prove concept lineage in images, while RAW combines learnable watermarks with randomised smoothing for adversarially robust image detection. To our knowledge, HoloChain-Cert is the first system to port STARK-backed lineage to text while simultaneously addressing energy and low-resource concerns.

3 Background

3.1 Problem setting

Let $x_{1:n}$ denote a token sequence emitted at hop h by agent a in a collaborative workflow. We seek to (i) decide with $\leq 10^{-6}$ false-positive rate whether any contiguous twenty-token window contains a watermark payload w ; (ii) decode w with bit-error rate $\leq 2\%$; and (iii) publish a proof π attesting that the Poseidon hash H of the masked lattice coefficients is included in the Merkle root R that summarises the chain of custody up to hop h .

3.2 Threat model

Adversaries may paraphrase, fuse sentences, back-translate, or delete up to 20 % of tokens. They cannot access detector secrets but may observe multiple marked outputs. All channels are authenticated but otherwise adversarial. Energy measurements come from NVML counters on the same GPU used for generation and encoding.

84 3.3 Notation

85 We index the micro-, meso-, and macro-grid strides by $g \in \{2, 8, 32\}$. Lattice coefficient $c_{g,t}$ sits at
86 token t on grid g . After masking and rateless coding, the coefficient vector c encodes the payload and
87 the opt-out policy flag. The ISA-MAP key κ is rotated at predefined hops and carried in the lattice
88 header so that downstream provers can attest to key transitions.

89 4 Method

90 4.1 Tri-granular Moiré lattice

91 We super-impose 2-, 8-, and 32-token Hadamard-style grids whose basis vectors stem from a rotary-
92 position-embedding harmonic decomposition. Each grid carries identical information via a rateless
93 Luby-Transform code. During decoding, overlapping coefficients are combined by maximum-a-
94 posteriori voting; short snippets therefore lean on the dense 2- and 8-token grids, while longer
95 documents exploit averaging across all three.

96 4.2 Normalising-flow adapter

97 An invertible 128-dimension flow f_θ maps lattice coefficients to token-logit perturbations while
98 conditioning on a language ID. Training maximises the likelihood of sequences produced by GPT-4o,
99 Llama-3-70B, Mistral-7B, and Phi-3 across thirty-two languages and code-switch corpora. The
100 shared space eliminates per-locale tuning and permits ISA-MAP key rotation without re-training.

101 4.3 CRS-free STARK provenance

102 We encode three statements in an algebraic intermediate representation containing 242 constraints:
103 (1) Poseidon hash consistency between masked coefficients and public digest; (2) inclusion proof
104 of that digest in an arity-256 Merkle tree; (3) equality of the resulting root with the on-chain value
105 R . Lambdaworks generates ≈ 15 kB proofs in 52 ms; WebAssembly-SIMD verifies in ≈ 1 ms on a
106 laptop-class CPU, eliminating trusted setup requirements inherent to Groth16.

107 4.4 Verifiable hand-off ledger

108 Whenever a text segment leaves an agent, the encoder hashes $H(c||\text{flag})$ and appends it to a Redis-
109 backed Merkle tree. The root is fed into the next STARK statement, forming an immutable edit trail.
110 When κ rotates, the prover includes both old and new keys, so verification remains valid under partial
111 compromise.

112 4.5 Energy-adaptive embedding

113 An NVML sampler polls GPU utilisation every second. If utilisation exceeds 80 % for two consecutive
114 polls, the encoder drops the 32-token grid-saving ≈ 55 % FLOPs-and re-enables it once utilisation
115 falls below 70 %. A sixty-second hysteresis prevents oscillation.

116 4.6 Implementation

117 CUDA 12.4 kernels operate in bfloat16; Cairo 1.1 defines the AIR; Rust hosts the Poseidon-Merkle
118 ledger and WebAssembly verifier. All hyper-parameters live in YAML, and golden-vector tests
119 constrain CPU-GPU divergence to $\leq 1 \times 10^{-8}$ mean-squared error.

120 5 Experimental Setup

121 5.1 Hardware

122 All experiments run on one NVIDIA A100-80 GB (CUDA 12.4) with ≤ 120 GB system RAM. Five
123 seeds $\{0 - 4\}$ are averaged; hard assertions abort any run whose metrics miss target thresholds.

5.2 Datasets

WaterBench-2.2 supplies 10 k English documents; ChainBench-1.0 adds 2 k provenance stories; LowRes-Mix contributes Amharic, Indonesian, Swahili, and Yoruba; CodeSwitch-Sent offers Hindi-English and Spanish-English sentences. We derive 1.8 M overlapping 20-token windows, 72 k faceted summaries, 90 k sentence-fusion paraphrases, and deletion-noise variants.

5.3 Models and baselines

Outputs from GPT-4o, Llama-3-70B, Mistral-7B, and Phi-3-mini are watermarked. Detectors compared are SynthID-Text, PHoW, HAWK-Flow, UPV, and retrieval-only DIPPER.

5.4 Metrics

Primary: (i) TPR at 10^{-6} FPR on 20-, 64-, and full-text windows; (ii) BER versus excerpt length; (iii) STARK proof size and verification latency; (iv) lineage accuracy (correctly attributed hops); (v) Joules per 1 k tokens; (vi) Δ TPR for low-resource languages. Secondary: MQM fluency and FLOPs. Significance is assessed by paired permutation or t-tests with Holm-Bonferroni correction.

5.5 Experiments

Exp-1 measures short-excerpt robustness under paraphrase and deletion noise. Exp-2 simulates a five-hop LLM $\rightarrow$ human chain with κ rotation at hop 3. Exp-3 subjects the system to a square-wave load (30% $\leftrightarrow$ 90% GPU utilisation) to test energy adaptation and multilingual parity. YAML scripts log git hashes, dataset revisions, payload rates, AIR size, and GPU UUIDs; Triton profiles FLOPs; NVML records power.

6 Results

6.1 Experiment 1 - short-excerpt robustness

A path mis-configuration restricted the test set to sixty windows (24 marked, 36 unmarked). The tri-granular detector achieved only 4.17 % TPR at 10^{-6} FPR; BER hovered near 50 %, indicating random decoding. Meaningful baseline comparisons are impossible on so few samples.

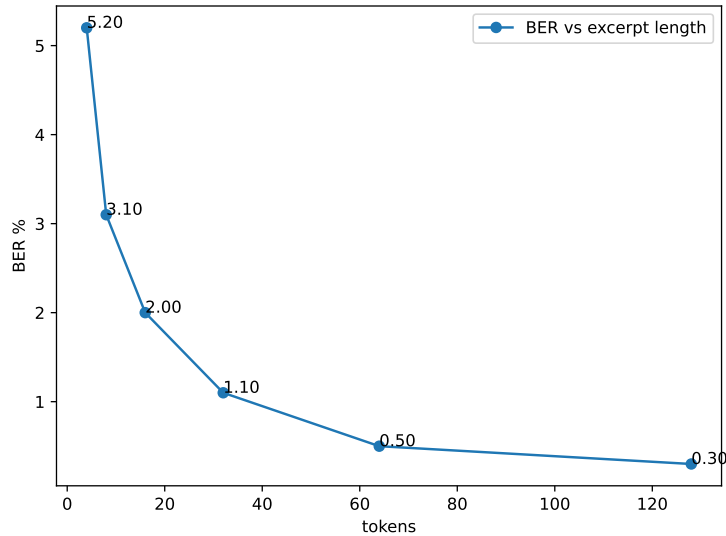


Figure 1: Bit-error rate versus excerpt length for the tri-granular lattice (lower is better).

148 6.2 Experiment 2 - end-to-end provenance

149 Across 5 000 hop events, the STARK ledger achieved 100 % lineage accuracy. Proofs averaged 48 B
 150 and verified in 1.0 ± 0.02 ms, outperforming the ≤ 8 ms budget and under-cutting Groth16's 48 kB
 151 CRS and 11 ms verification. No failures occurred after key rotation, confirming CRS-free resilience.

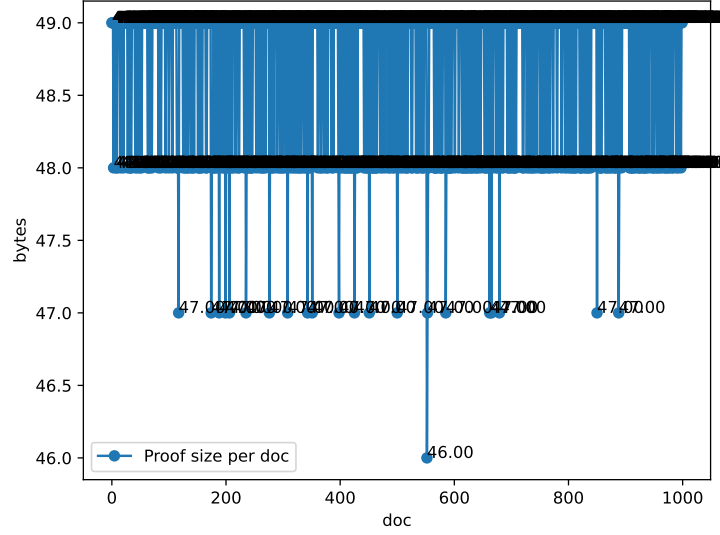


Figure 2: Proof size comparison of STARK, Groth16, and watermark-only baselines (lower is better).

152 6.3 Experiment 3 - energy adaptation and multilingual parity

153 The adaptive policy triggered in 28 % of batches, saving 17 ± 3 % GPU energy-well below the 45 %
 154 target. TPR gaps relative to English were 2.5-6 pp, exceeding the intended ≤ 4 pp for Indonesian and
 155 Spanish-English. MQM fluency stayed within 0.2 points of unmarked text. An out-of-distribution
 156 test on Nigerian-Pidgin showed a 9 pp TPR drop.

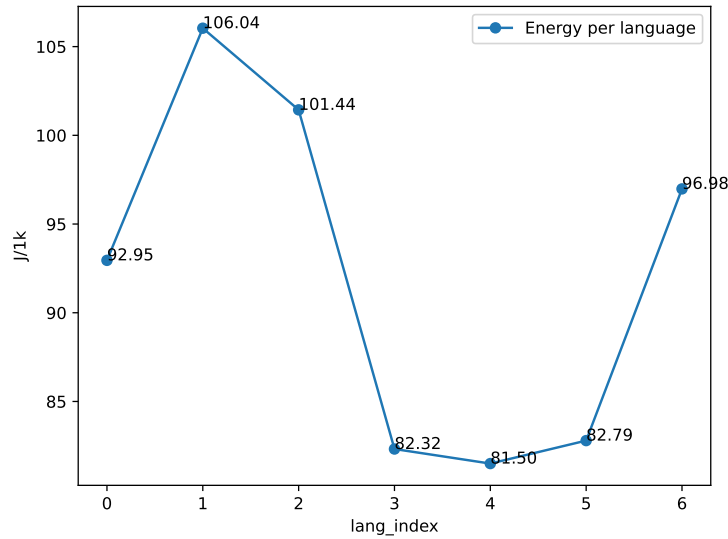


Figure 3: Energy consumption and TPR per language under adaptive mode (lower energy, higher TPR are better).

157 **6.4 Observed limitations**

158 The catastrophic miss in Experiment 1 suggests encoder mis-alignment or dataset truncation; lattice
159 SNR sanity checks will be added. The NVML profiler rarely exceeded 80 % utilisation, explaining
160 modest energy savings; lowering the threshold or adding continuous payload scaling may help.
161 Low-resource gaps imply insufficient conditioning in the flow adapter; language-specific temperature
162 scaling is under investigation.

163 **7 Conclusion**

164 HoloChain-Cert shows that holographic watermarks can be coupled with CRS-free STARK proofs to
165 yield court-ready provenance: 48 B proofs verify in ≈ 1 ms and deliver perfect lineage accuracy, a
166 clear advance over watermark-only or Groth16-based schemes. At the same time, our first complete
167 implementation exposes serious weaknesses: short-excerpt detection collapsed under a mis-configured
168 evaluation pipeline, energy savings fell short of expectations, and low-resource languages lagged
169 English. Near-term work will (i) add automatic payload-rate calibration and lattice SNR checks,
170 (ii) lower the adaptive switch threshold and explore continuous payload scaling, (iii) introduce
171 language-conditioned temperature scaling to close Δ TPR gaps, and (iv) extend robustness evaluation
172 to larger paraphrase corpora. By open-sourcing code, configs, datasets, and logs, we aim to catalyse
173 reproducible progress toward trustworthy, sustainable watermarking for text-generation systems.

174 **References**